

Sentiment Prediction of Text Review

A Project Report

Submitted in partial fulfilment of the
requirements for the award of the Degree of

BACHELOR OF SCIENCE (DATA SCIENCE)

By

Kapish Ashok Shukla

Seat Number: -----

Under the esteemed guidance of

Ms. Kanchan Taksale

Assistant Professor, Department of Information Technology

And Data Science



DEPARTMENT OF INFORMATION TECHNOLOGY AND DATA SCIENCE

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

(Affiliated to University of Mumbai)

MUMBAI, 400 037

MAHARASHTRA

2024 - 2025

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

(Affiliated to University of Mumbai)

MUMBAI-MAHARASHTRA-400037

DEPARTMENT OF INFORMATION TECHNOLOGY

AND DATA SCIENCE



CERTIFICATE

This is to certify that the project entitled, "**Sentiment Prediction of Text Review**", is bonafied work of **KAPISH ASHOK SHUKLA** bearing Seat No: 23315A0040 submitted in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE in DATA SCIENCE from University of Mumbai.

Internal Guide

Coordinator

Internal Examiner

External Examiner

Date:

College Seal

Principal

ABSTRACT

The project "Sentiment Prediction of Text Review" is aimed at developing an automated system to classify the movie reviews using machine learning. A labeled dataset of 50,000 reviews forms the foundation for training and evaluating models. Text preprocessing and feature engineering methods includes TF-IDF and Count Vectorization which were applied to transform unstructured text into meaningful numerical representations.

Multiple classifier like Logistic Regression, Naive Bayes, Random Forest and Support Vector Classifier, were assessed for making the model learn to predict classes, alongside an ensemble Voting Classifier that integrates their strengths. Model evaluation was conducted using accuracy and F1-score to ensure balanced performance across classes.

The project streamlines sentiment analysis by minimizing manual effort, improving efficiency and offering actionable insights to stakeholders in the movie industry. Key objectives include enhancing brand reputation management, uncovering opportunities for service improvement and supporting data-driven decision-making. By intelligently capturing sentiment behind customer feedback, the system demonstrates the practical value of sentiment analysis in improving business processes and audience satisfaction within the entertainment sector.

ACKNOWLEDGEMENT

I want to thank everyone who helped me during my semester project. Without their support and advice, I could not have completed it.

I also appreciate Vidyalankar School of Information Technology for giving me this opportunity. The friendly learning environment encourages innovation and research. I look forward to using the resources and support from the institution during my project.

First, I want to express my deep gratitude to my project guide, Prof. Kanchan Taksale. Her advice, encouragement and feedback were vital to my progress. She has been a great mentor and a constant source of inspiration. Her expertise in natural language processing and machine learning helped me tackle challenges and improve the quality of my work.

I want to thank all the faculty members at Vidyalankar School of Information Technology. Their teaching and guidance have greatly impacted my learning. I especially appreciate the administrative staff for their help and my fellow students for their valuable discussions and support throughout this journey.

DECLARATION

I declare that the project titled “Sentiment Prediction of Text Review,” completed at Vidyalankar School of Information Technology, has not been submitted to any other universities for any degree. To the best of my knowledge, aside from myself, no one has submitted it to any other university.

This project is done to partially fulfill the requirements for the **BACHELOR OF SCIENCE (DATA SCIENCE)** degree and it will be submitted as the final semester project for our curriculum.

Name and Signature of the Student

TABLE OF CONTENTS

Chapter 1 Introduction	9
1.1 Background.....	9
1.2 Objectives.....	18
1.3 Purpose, Scope, Applicability (Feasibility Study)	20
Chapter 2 Dataset Description.....	23
2.1 Introduction	23
2.2 Data Source	23
2.3 Dataset Characteristics	24
2.4 Data Features	24
2.5 Dataset Usage	25
2.6 Limitations of the Dataset.....	25
Chapter 3 Methods and Algorithms	27
3.1 Overview of Methods Used	27
3.2 Models Used	28
3.3 Model Evaluation.....	30
3.4 Hyperparameter Tuning.....	30
3.5 Challenges and Future Directions	30
Chapter 4 Project Analysis	31
4.1 Data Access and Structure.....	31
4.2 Data Preprocessing and Feature Engineering.....	33
4.3 Data Distribution and Exploratory Analysis	35
4.4 Model Performance	37
4.5 Error Analysis	38
4.6 Challenges and Limitations	39
4.7 Conclusion.....	39
Chapter 5 Final Results.....	40
5.1 Model Performance Evaluation.....	41
5.2 Model Limitations and Challenges	42
Chapter 6 Conclusion and Future Scope.....	44
6.1 Conclusion.....	44
6.2 Future Scope	44
6.3 Closing Remarks	46
.References.....	46

Bibliography	47
Glossary	49
Summary	51
Further Reading	52
Plagiarism Report	54

List of Figures

Figure 1: Sample rows of IMDb dataset	31
Figure 2: Python code for preprocessing text reviews.	32
Figure 3: TF-IDF vector representation.	33
Figure 4: Distribution of reviews in dataset.	34
Figure 5: Word cloud of positive reviews.	35
Figure 6: Word cloud of negative reviews.	35
Figure 7: Accuracy comparison of different models.	Error! Bookmark not defined.

Chapter 1 Introduction

The *Sentiment Prediction of Text Review* project explores sentiment analysis in the movie industry using machine learning. Its main goal is to create an automated system that classifies reviews as positive or negative. Movie reviews significantly impact audience perception, brand reputation, and business decisions in the entertainment sector [2]. Traditionally, teams evaluated sentiment in reviews manually, which was time-consuming, resource-intensive, and subject to human bias. With the growth in Natural Language Processing (NLP) and machine learning [1], [2], this project attempts to automate sentiment recognition, reducing the need for manual work while providing accurate and scalable predictions.

This project depends on a labelled dataset containing 50,000 movie reviews [5]. Each review is marked for its sentiment, thereby paving the way for employing supervised learning approaches. To bring the data into a form the model can understand, feature extraction techniques like Count Vectorization and Term Frequency–Inverse Document Frequency (TF-IDF) were used [4]. A set of machine learning models, which comprises Logistic Regression, Random Forest Classifier, Naïve Bayes, and a Voting Classifier, were trained and evaluated to efficiently predict sentiment [1], [3], [11].

Throughout the project, several challenges emerged, including the complexities of natural language such as sarcasm, ambiguous expressions, and domain-specific vocabulary. Handling a large dataset also posed challenges in maintaining consistent accuracy and generalization. Moreover, finding a balance between interpretability and performance was crucial, as some high-performing models often functioned like “black boxes” [2], [12].

1.1 Background

1.1.1 Overview

Sentiment analysis, also called opinion mining, has become one of the most important applications of natural language processing and machine learning [2], [13]. At first, understanding human opinions was done manually. This required a lot of time and effort to be invested by individuals or teams. However, with the rise of large digital platforms and the surge of online reviews, traditional methods have become inefficient and hard to maintain [2]. Over the period of time, sentiment analysis has progressed

from basic keyword matching to more sophisticated machine learning techniques that can understand the relative importance of words, their meaning, and deeper patterns in text [1], [3], [4]. Businesses in various fields, whether retail or entertainment, have increasingly used sentiment prediction systems to make better decisions, protect their reputation, and improve customer engagement based on the credible information provided by data [2], [13].

In the movie industry, the importance of sentiment analysis is particularly clear. Movie reviews have a huge impact on how audiences see a film, how much money it makes, and the reputation of production companies and streaming platforms [2]. A single wave of positive or negative sentiment can directly influence commercial success, making accurate sentiment analysis a vital tool [2]. The main goal of this project, *Sentiment Prediction of Text Review*, is to use machine learning techniques to analyse and classify sentiments in large movie review datasets [5]. By automating this, businesses can improve customer experience, spot areas for improvement, and support data-driven decision-making in a competitive entertainment market [2], [13].

1.1.2 Rationale for choosing this topic

The choice of this project is inspired by the significant influence of audience sentiment, which plays a key role in the success and perception of movies [2], [13]. With the exponential growth of online platforms such as IMDb, Rotten Tomatoes, and streaming service review sections, vast amounts of unstructured text data are generated daily by moviegoers in the form of reviews containing their opinions [2]. These reviews then become a goldmine for uncovering audience preferences, identifying recurring themes, and predicting overall satisfaction [13]. However, manually analyzing such large volumes of feedback is time-consuming, resource-intensive, and subject to human bias [2].

This project aims to address this challenge by leveraging Natural Language Processing (NLP) and machine learning to automate the process of evaluating movie reviews followed by determination of sentiment [1], [3], [11]. Automating sentiment detection not only reduces manual effort but also provides production houses, distributors, and streaming platforms with actionable insights at scale, with a decent level of credibility [2], [12]. For example, studios can better understand what aspects of a film resonate most with audiences—

whether it be storytelling, acting, music, or visual effects—and can use these insights to refine marketing strategies, content development, and customer engagement [13].

The project is also motivated by the broader industry trend of combining data science into entertainment decision-making. Tools that analyse real-time audience sentiment have already begun to influence content recommendation systems, advertising strategies, investment decisions, and even story development tailored to maximum audience liking [2], [12]. By focusing on sentiment prediction in movie reviews, this project seeks to bridge the gap between raw audience feedback and data-driven, business-oriented decision-making in the entertainment sector [13].

1.1.3 Significance

The significance of this project lies in its ability to transform how movie reviews are understood and consumed by the business in the film industry to generate meaningful insights [2], [13]. By automating sentiment prediction, it becomes possible to analyse vast volumes of audience feedback quickly and consistently, thereby saving time for teams and allowing them to focus on more important tasks [2]. This enables stakeholders to identify strengths such as impactful storytelling or outstanding performances, qualities of an actor/actress, while also recognizing recurring pain points that could hinder reception [13].

For audiences, this project contributes to more personalized recommendations and engaging viewing experiences, as platforms can better align with user preferences, thereby giving audiences the feeling of being heard [2]. Additionally, researchers and industry analysts can use these insights to conduct studies of the gradual development of viewer trends and cultural responses to cinema [13]. Beyond immediate business value, the project highlights the potential of basic Natural Language Processing techniques to fill the gap between raw unstructured data and actionable decision-making in the entertainment sector [2], [12].

1.1.4 Challenges

The very first challenge in this project is the complexity and variability of movie reviews. Unlike structured data, reviews are written in informal language which makes them unstructured data, often containing slang, abbreviations, emojis, or even sarcasm,

which makes it difficult for basic NLP methods to capture the true sentiment accurately [2], [12]. The presence of biased or fake reviews adds another layer of noise, further complicating the process of building a reliable prediction system [13]. In addition, datasets are often imbalanced, with more positive reviews than negative ones, due to which the model predicts positive reviews much more accurately than the negative ones [2].

As the size of the dataset increases and sentiment classes expand beyond a simple positive-versus-negative framework to include neutral or multiple sentiment categories, the difficulty of maintaining accuracy grows [13]. Basic approaches such as Bag-of-Words or TF-IDF vectorization struggle to detect subtle cues in language context, making sarcasm, mixed sentiments, or implicit criticism particularly difficult to classify [4].

From a business perspective, integrating an automated system that predicts sentiment into decision-making pipelines also poses difficulties. Film production houses and streaming platforms not only require accurate results but also require insights that can be mapped to actionable strategies, such as refining marketing campaigns or tailoring recommendations [2], [13]. Ensuring that these models remain interpretable, scalable, and adaptable to the fast-changing dynamics of audience conversations poses a significant challenge for industry adoption [12].

1.1.5 Justification

The need for this project arises from the fact that the volume of movie reviews is growing across online platforms and the time consumption of extracting meaningful insights from them manually [2], [13]. With audience discussions shaping the reputation and commercial performance of films, relying only on human analysis is inefficient, prone to oversight, and time-consuming [2]. By building an automated sentiment prediction system, the film industry can track and understand audience reactions more efficiently [13]. This reduces manual work and reveals valuable insights into viewer preferences and concerns [2]. For audiences, such a system contributes to

more relevant recommendations and an enhanced movie-watching experience through platforms that better understand their sentiments [12], [13].

The technological aspect of this project makes sense, given the success of Natural Language Processing in areas like customer feedback analysis and product reviews [2], [12]. By using these methods in the movie industry, the system can effectively sort unstructured text into sentiments [1], [3], [4]. This helps identify what excites or disappoints viewers [13]. Practical applications include allowing production houses to create more targeted marketing campaigns, assisting streaming platforms in building recommendation systems, and helping industry stakeholders make decisions based on data that matches audience expectations [2], [13].

1.1.6 Historical Methods

Overview of Historical EEG Interpretation -

Historically, how we interpret movie reviews has changed with the growth of cinema. In the early days of the film industry, people shared feedback mainly through word-of-mouth. A movie's reputation relied heavily on local conversations and personal recommendations [13]. Eventually, newspaper and magazine critics became the main voices interpreting films. They provided structured but limited views that represented only a small group of experts. While these reviews were influential, they often did not reflect the varied opinions of the broader audience [2].

With the rise of the internet and many online platforms, reviews changed from being mainly driven by critics to being driven by audiences. Websites like IMDb, Rotten Tomatoes, and social media let millions of viewers share their thoughts [2]. However, trying to understand such large amounts of unorganized text by hand turned out to be time-consuming and inconsistent [12]. The absence of systematic ways to analyse this expanding collection of user reviews meant that valuable insights, such as common themes in audience preferences or general dissatisfaction with certain elements, were often missed [13]. This change shows the historical challenge of moving from limited, critic-focused feedback to extensive, unstructured audience feedback that needs technology for proper analysis [2], [12].

Challenges of Historical Methods

The main challenge of historical methods in interpreting movie reviews is their reliance on film critics, word-of-mouth, and other manual ways of gathering audience opinions. While critics influenced public views, their opinions reflected only a small part of viewers and often included biases [13]. Word-of-mouth and fan letters also had no structure and limited reach, so they could not capture the wide range of feelings from large and diverse audiences [2]. This limited and scattered feedback made it hard for the film industry to truly grasp the overall feelings of moviegoers [13].

Before online platforms emerged, there was no standard method to document or understand audience reactions on a large scale. Interpretations varied across regions, media outlets, and personal views, leading to major inconsistencies in how sentiments were recorded or perceived [2]. As online reviews increased with the rise of digital

platforms, these traditional manual methods fell short [12]. The quick surge in unstructured text data made human evaluation more difficult and time-consuming [2], [12]. This situation highlighted the need for automated methods to find patterns, measure audience sentiment, and turn feedback into practical insights for stakeholders [13].

1.1.7 Modern Advancements

Innovations in Technology

Recent technological advancements in Natural Language Processing (NLP) have made it easier to analyse and interpret text reviews, which used to require a lot of manual work [2], [12]. Basic techniques like tokenization, Bag-of-Words representations, TF-IDF scoring, and lexicon-based sentiment analysis have allowed researchers and industries to change unstructured audience reviews into structured insights [4], [13]. These tools help extract key features from text automatically, lowering the need for manual reading and interpretation of large numbers of reviews [2], [12].

Machine learning methods used on these textual features have improved the ability to classify sentiments as positive, negative, or neutral with good accuracy [1], [3], [11]. What once took a lot of human effort can now be done efficiently, allowing for the processing and understanding of thousands of movie reviews in real time [2]. These NLP-based techniques show a modern change in how the entertainment industry handles audience feedback. They connect raw textual data to useful business insights for filmmakers, distributors, and streaming platforms [13].

Advantages of Modern Techniques

The benefits of modern techniques compared to older methods are clear in the area of movie review analysis. Unlike manual interpretation, which takes a lot of time and is limited in scope, Natural Language Processing and machine learning approaches can quickly process large amounts of audience reviews [2], [12]. This ability to scale allows for capturing not just a few opinions, but thousands of different audience sentiments across platforms. This ensures a better understanding of how a film is received [13]. The accuracy also improves since these techniques lessen the impact of individual bias that often affected traditional critic-based reviews [2], [13].

Another key advantage is the ability to perform near real-time sentiment monitoring. This is especially valuable during the important days around a film's release [13]. Production houses, distributors, and streaming platforms can track shifts in audience perception almost

instantly, allowing for agile marketing strategies, better audience engagement, and timely interventions if negative sentiments arise [2]. Moreover, these techniques reveal patterns in unstructured text. They help stakeholders pinpoint specific strengths or audience concerns, such as appreciation for performances or dissatisfaction with the storyline [12], [13]. By combining speed, scale, and practical insights, modern sentiment prediction techniques provide the movie industry a significant improvement over traditional methods of review interpretation [2], [13].

1.2 Objectives

Objective 1: Reduce the Workload on Manual Teams

The main goal of this project is to automate how we identify sentiment in movie reviews. This will lessen the workload for teams that currently handle this task manually. Interpreting a large number of reviews by hand takes a lot of time and can lead to inconsistencies and human bias. By using automated sentiment prediction, film production companies and streaming services can make their operations more efficient. This approach saves time and resources while providing more reliable results.

Objective 2: Retain Brand Reputation

Another key goal is to help keep and protect brand reputation in the movie industry. Online reviews greatly affect how audiences see films and streaming platforms. By quickly spotting and analyzing negative feedback, businesses can respond early to problems before they get worse. This approach helps maintain their image and build trust with their audience.

Objective 3: Identify Areas for Improvement

This project aims to highlight areas needing improvement, whether in movie content or in the services provided by streaming platforms. Sentiment prediction allows industry stakeholders to see where audiences are unhappy with specific aspects, such as storyline, casting or streaming quality. These insights create focused chances for improvement, helping filmmakers and platforms refine their offerings to better match audience expectations.

Objective 4: Enable Data-Driven Decision Making

A central goal is to help industry stakeholders make decisions based on data instead of just relying on intuition or random feedback. Automated sentiment analysis changes unstructured text reviews into organized, useful insights. This allows production houses, distributors and streaming services to shape their marketing strategies, content investments and release planning according to real audience reactions.

Objective 5: Enhance Customer Experience

Finally, this project aims to improve customer experience by identifying and addressing audience pain points more effectively. By systematically reviewing feedback, businesses can understand what delights or frustrates viewers and take actions to improve. This not only boosts audience satisfaction but also builds stronger long-term loyalty to filmmakers, studios and streaming platforms.

1.3 Purpose, Scope, Applicability (Feasibility Study)

1.3.1 Purpose

The goal of this project, *Sentiment Prediction of Text Review*, is to create a system that can automatically analyse and classify sentiments from movie reviews using Natural Language Processing techniques. The project seeks to lessen the manual work needed to understand large amounts of audience reviews while improving accuracy, consistency and scalability in sentiment evaluation. By providing near real-time insights, the system will help production companies, distributors and streaming services monitor audience opinions effectively and protect their brand reputation.

This project focuses on business use by giving stakeholders actionable insights. These insights guide data-driven decisions, support targeted marketing strategies and identify areas for improvement in films or services. At the same time, by addressing key audience issues mentioned in text reviews, the system helps improve the overall customer experience. In combining these perspectives, the project seeks to bridge the gap between unstructured audience feedback and practical business strategies in the entertainment industry.

1.3.2 Scope

The scope of this project covers the development of an NLP-based sentiment prediction system that can analyse and classify movie reviews. The system will:

- Preprocess textual reviews by performing steps such as tokenization, stop-word removal and feature representation using approaches like Bag-of-Words and TF-IDF.
- Apply basic NLP and machine learning methods to classify reviews into two distinct sentiment categories: positive and negative.
- Automate the extraction of key insights from large volumes of unstructured text, reducing the reliance on manual review interpretation.
- Provide stakeholders in the movie industry such as production houses, distributors and streaming platforms with actionable insights to safeguard brand reputation, identify areas for improvement and make data-driven decisions.

- Enable near real-time monitoring of audience feedback, especially during critical periods such as film releases, allowing businesses to respond rapidly to audience sentiment.

The project will be limited to analyzing **text-based movie reviews only** and will not address video reviews, audio reviews or reviews outside the film industry domain.

1.3.3 Applicability

This system is applicable in a variety of domains within the movie industry and audience engagement:

- **Product and Service Improvement:** Streaming platforms and studios can leverage insights from sentiment classification to understand specific aspects of dissatisfaction, such as storyline, acting performances or streaming quality. This supports the objective of identifying areas for improvement in films and services.
- **Data-Driven Decision Making:** The system aids stakeholders in using structured insights from unstructured text reviews to make more informed business choices. Whether guiding investment decisions, shaping marketing strategies or forecasting audience reception, this directly fulfills the objective of enabling data-driven decisions.
- **Customer Experience Enhancement:** By capturing audience pain points highlighted in reviews, the system allows platforms to make adjustments that improve user satisfaction. This supports the goal of improving the overall customer experience and building long-term loyalty.
- **Audience Engagement:** For viewers, the system turns collective feedback into useful improvements. This can mean better film quality or more relevant content recommendations on platforms. This goal helps meet both audience needs and business goals at the same time.

1.3.4 Significance of the System

The importance of this system is in its capability to connect unstructured audience reviews with useful business insights using modern Natural Language Processing techniques. By automating sentiment analysis for movie reviews, the system lessens the workload on manual teams and also enables the industry to scale its

understanding of audience perceptions with greater accuracy and efficiency. This helps production houses, distributors and streaming platforms protect their brand reputation, direct investments and create effective marketing strategies.

Beyond business benefits, the system also plays an important role in improving the viewer experience. By identifying audience challenges and positive feelings more effectively, filmmakers and platforms can make changes that better meet audience expectations. This builds a stronger relationship between the audience and the business, ensuring lasting trust and loyalty. Moreover, the system shows the practical use of technology in the entertainment industry. It illustrates how automated sentiment prediction can turn subjective reviews into actionable plans that help both industry stakeholders and audiences.

1.3.5 Technical feasibility

The technical feasibility of this project relies on the availability of established Natural Language Processing (NLP) and machine learning techniques [2], [12]. Methods like tokenization, text normalization, Bag-of-Words, and TF-IDF make it easy to preprocess and extract features from large amounts of unstructured reviews [4], [12]. Supervised learning algorithms, such as logistic regression, Naïve Bayes, and SVM, have been widely used in sentiment classification tasks [1], [3], [11]. They can effectively manage binary classification of movie reviews with high accuracy [13].

Challenges like spelling errors, slang, and inconsistencies in review text can be managed through preprocessing steps [12]. This ensures data quality and model reliability. Using the IMDb dataset offers a consistent and scalable resource for training and testing [5]. This makes experimentation and validation practical. With the growing computational power for model training and the scalability of current algorithms, the system can handle large datasets efficiently and provide reliable performance [2], [11]. This ensures the project's technical feasibility.

Chapter 2 Dataset Description

2.1 Introduction

The dataset for this project comes from the IMDb Movie Reviews Dataset, which is publicly available on Kaggle [5]. This dataset is widely used in research related to natural language processing and sentiment analysis because of its large size and high-quality labeling [2], [13].

For this project, the dataset includes 50,000 text reviews, each labeled with a binary sentiment classified as either positive or negative [5]. It has two main columns: one with the raw text of user movie reviews and the other with the corresponding sentiment label. Unlike some other review datasets, this version does not have extra metadata like reviewer information, timestamps, or ratings. This keeps the focus on the textual content and its sentiment [13].

The dataset offers a balanced distribution of positive and negative reviews, making it a reliable standard for training and testing machine learning models in sentiment prediction tasks [5]. Its simplicity and size make it especially suitable for exploring methods that range from traditional machine learning models to deep learning approaches, aiming to classify sentiment from unstructured text [2], [6], [7], [8].

2.2 Data Source

The IMDb Dataset of 50K Movie Reviews comes from a publicly available dataset on Kaggle [5]. It is well known and widely used for natural language processing tasks, particularly in sentiment analysis [2], [13]. The dataset can be accessed through the Kaggle platform at the following link: <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews> [5]. It includes 50,000 movie reviews divided equally, with binary sentiment labels showing either positive or negative sentiment [5].

The dataset is free to download under Kaggle's terms and conditions, which require users to register for a Kaggle account. Each review contains only raw text and its sentiment label without any extra metadata, keeping the dataset focused on textual content [13]. For the purpose of this project, the raw review texts and binary sentiment labels were employed without initial alteration; however, standard text preprocessing

steps such as removal of HTML tags, stop word elimination, and tokenization were applied prior to model development to enhance data quality and support effective sentiment prediction [2], [12].

2.3 Dataset Characteristics

The IMDb Dataset containing Movie Reviews consists of the following characteristics:

1. **50,000 reviews:** The dataset includes 50,000 movie reviews, evenly divided into 25,000 positive and 25,000 negative sentiment labels. This setup supports balanced binary classification [5].
2. **Text and Label Columns:** Each record includes a raw textual review and its corresponding binary sentiment label. There is no extra metadata, which keeps the dataset focused on sentiment classification [5], [13].
3. **Review Length:** Reviews can be short captions or detailed stories. They show a wide variety of text styles and word choices, reflecting the diversity of natural language in real-world user-generated content [2], [12].

This dataset offers a varied collection of movie reviews from the IMDb platform. It shows both positive and negative audience feelings about films, making it a reliable benchmark for sentiment analysis research [5], [13]. The data is mostly text-based and is used for sentiment prediction after standard preprocessing steps such as tokenization, stop word removal, and normalization to prepare the text for machine learning modeling [2], [12].

2.4 Data Features

The dataset has several key textual features that are important for predicting sentiment [2], [13]. The most significant features used in this project are:

- **Review Text:** The raw text of each movie review is the main input for sentiment analysis. This includes all the words, phrases, and sentence structures that users write to share their opinions about movies [2], [12].
- **Sentiment Label:** A binary classification label shows the overall sentiment of the review. It is categorized as either positive or negative. This label is the target variable used for training and evaluating the model [5], [13].

These features are essential for creating models that understand and classify the sentiment expressed in textual reviews [2], [13].

2.5 Dataset Usage

The dataset is publicly available on the Kaggle platform. Users can download the full set of 50,000 movie reviews along with their binary sentiment labels. The dataset is organized well, making it easy for researchers and practitioners to use it for training and evaluating sentiment prediction models.

While the dataset does not include extra metadata, the text reviews and sentiment annotations are organized into columns. This makes it easy to integrate into natural language processing workflows. Users can use this dataset for different applications, such as binary sentiment classification, text mining and model testing. Standard preprocessing and feature extraction techniques are usually applied before inputting the data into machine learning algorithms.

2.6 Limitations of the Dataset

- **Limited Sentiment Classes:** The dataset includes only two sentiment labels, positive and negative. It does not cover neutral or mixed sentiments. This restriction limits how detailed the sentiment prediction can be.
- **Lack of Additional Metadata:** The dataset does not include any extra information like reviewer details, timestamps or movie genres. This information could have been useful for better context analysis or developing new features.
- **Fixed Review Source:** All reviews come from the IMDb platform. This may create platform-specific biases and limit how models trained only on this dataset can be applied to other areas or types of reviews.
- **Balanced Dataset Constraints:** The mix of positive and negative reviews helps with training, but it doesn't represent how data looks in the real world, especially with only

such a limited number of reviews. Data can include sarcasm or jargons in real-world, which might impact how strong the model is.

Chapter 3 Methods and Algorithms

This chapter gives an overview of the methods and algorithms used in this project to predict sentiment from text reviews. The main steps in the process include data preprocessing, feature extraction, model selection, training, and evaluation [2], [12]. The project mainly uses machine learning techniques to classify text reviews into two sentiment categories: positive or negative [13].

The chapter discusses the different algorithms examined, from traditional models like Logistic Regression, Naïve Bayes, and Support Vector Machines (SVM) [1], [3], [11], to more complex deep learning methods such as Word2Vec, GloVe, and BERT [6], [7], [8]. It outlines the overall approach taken for successful sentiment prediction, highlighting how these methods transform raw textual data into structured insights that can be used for classification [2], [13].

3.1 Overview of Methods Used

The main goal of this project is to predict how people feel about movie reviews using machine learning methods on text data. The following key steps were undertaken in the process:

1. **Data Preprocessing:** This step involved cleaning the raw text data by removing noise such as HTML tags, punctuation, and stop words. Tokenization and vectorization methods, such as TF-IDF, were applied to transform the text into numerical features that machine learning models can understand [2], [4], [12].
2. **Model Selection and Evaluation:** Multiple machine learning algorithms were examined to find the best method for sentiment classification, including Logistic Regression, Random Forest, Voting Classifier, and Naïve Bayes [1], [3], [11], [13]. The performance of each model was assessed using metrics such as accuracy, precision, recall, and F1-score, which provided a comprehensive evaluation of their effectiveness on the dataset [2], [13].
3. **Feature Selection:** Techniques like TF-IDF vectorization were used to extract relevant features from the text data. Dimensionality reduction and feature importance analysis were performed to identify the most significant features for the classification task [4], [12].
4. **Hyperparameter Tuning:** The top-performing models had their hyperparameters fine-tuned using grid search and cross-validation. This process aimed to improve predictive accuracy and ensure better generalization ability [11], [13].

3.2 Models Used

3.2.1 Logistic Regression

Logistic Regression is a widely used linear model for binary classification tasks. It estimates the likelihood of a review being positive or negative by fitting a logistic function to the input features [2], [11]. This model is valued for its simplicity, clear interpretation through feature coefficients, and efficient training, even with high-dimensional text data such as movie reviews [2], [13].

Advantages:

- Simple and quick to implement [11]
- Easy to understand, particularly with feature importance derived from coefficients [2]
- Performs well for linearly separable data [13]

Challenges:

- Assumes a direct linear relationship between features and classes, which may not always hold true for the complexities of textual data [2], [13]

3.2.2 Random Forest Classifier

Random Forest is an ensemble learning method that combines several decision trees and aggregates their results to improve classification performance [9]. It operates by training on different bootstrapped samples of the dataset and using random feature selection at each split, which reduces overfitting and increases generalization capability [9], [11]. In the context of sentiment prediction of text reviews, Random Forest leverages diverse subsets of features (such as word embeddings, TF-IDF scores, or other textual representations) to capture different aspects of the data and produce a more robust classification outcome [2], [13]. The collective decision-making process helps the model handle variability in text expressions and noise that is common in natural language data [2].

Advantages:

- Handles high-dimensional feature spaces well, which is useful for text data [9], [11]
- Reduces the risk of overfitting compared to single decision trees [9]
- Provides feature importance, enabling interpretability of which words or features are most influential in sentiment prediction [13]

Challenges:

- Computationally more expensive due to training multiple trees [9]
- May not perform as well as deep learning approaches on very large text datasets [6], [7], [8]
- The model can become less interpretable as the number of trees increases [9], [13]

3.2.3 Voting Classifier

Voting Classifier is an ensemble method that combines predictions from several different classifiers to reach a final decision [10], [11]. Instead of depending on just one model, Voting Classifier gathers outputs from base models such as Logistic Regression, Naïve Bayes, Support Vector Machine, or Random Forest. There are two common methods: **hard voting**, where the class with the majority vote is chosen, and **soft voting**, where the class probabilities predicted by the models are averaged, selecting the class with the highest probability [10].

For sentiment prediction of text reviews, Voting Classifier is useful because it merges the strengths of various models. This balance reduces bias and variance, improving overall prediction accuracy when dealing with complex textual data [2], [13].

Advantages:

- Combines strengths of multiple models, reducing the chance of relying on one weak classifier [10], [11]
- Provides improved accuracy and generalization across diverse review datasets [2], [13]
- Flexible, as it can integrate both simple and complex classifiers [11]

Challenges:

- Computationally more demanding since it requires training and maintaining multiple models [10]
- Performance highly depends on the choice and diversity of base classifiers [11]
- Can be less interpretable compared to single models, since final decisions come from combined votes [13]

3.2.4 Naïve Bayes

Naïve Bayes is a probabilistic classifier that applies Bayes' Theorem and relies on strong (naïve) independence assumptions between features [1]. It works particularly well when the features are conditionally independent. This method has been successfully applied to classification problems involving text, image, and audio data [2], [13]. In sentiment analysis, Naïve Bayes is widely used because of its simplicity, efficiency, and effectiveness in handling high-dimensional text data [1], [2].

Advantages:

- Fast and efficient, making it suitable for large-scale text classification [1], [11]
- Works well with small datasets and when features are independent [2], [13]

Challenges:

- Assumes independence among features, which may not always be realistic in textual data [2], [13]

- Performance can degrade when features are highly correlated, as is often the case in natural language [1]

3.3 Model Evaluation

Once the models were trained and the features extracted, their performance was evaluated based on their ability to classify the sentiment of text reviews. The assessment was carried out using **accuracy** and **F1-score**, as these metrics capture both the overall correctness and the balance between precision and recall [2], [11]. Accuracy provides a straightforward measure of how often the model predicts correctly, while the F1-score is particularly important in sentiment analysis because it balances precision and recall, ensuring that both positive and negative classes are treated fairly [2], [13].

This evaluation ensured that the models were not only accurate but also effective in handling the variability and potential imbalance often present in textual sentiment data [13]. By using these metrics, the project was able to compare different algorithms consistently and identify the approaches that generalized best to unseen reviews [11].

3.4 Hyperparameter Tuning

To improve the performance of the models, hyperparameter tuning was conducted, with particular focus on **Logistic Regression**. This process involved systematically adjusting the model's hyperparameters, such as the **regularization strength (C parameter)** and the **solver type**, to achieve the best accuracy [11]. Grid search combined with cross-validation was employed to evaluate different parameter combinations, ensuring that the model generalized well to unseen data [2], [13].

Hyperparameter tuning is a critical step in machine learning, as it allows models to balance bias and variance, avoid overfitting, and optimize predictive performance [2], [11]. By fine-tuning these parameters, the project ensured that Logistic Regression and other classifiers achieved robust and reliable results on the sentiment classification task.

3.5 Challenges and Future Directions

While significant progress was made in the sentiment prediction of text reviews, several challenges remain. These include handling **class imbalances**, which can bias models toward majority classes and reduce performance on minority classes [13]. Another challenge is managing **noisy and ambiguous user-generated content**, such as sarcasm, slang, and spelling errors, which complicate accurate sentiment detection [2], [12]. Additionally, maintaining robust performance across **diverse domains** (e.g., product reviews, social media, or news articles) remains difficult, as models trained on one dataset often struggle to generalize effectively [13].

Future directions involve incorporating **transformer-based deep learning models** such as BERT, which have demonstrated state-of-the-art performance in sentiment analysis and other NLP tasks [8]. Improving **feature representations** through contextual embeddings (e.g., Word2Vec, GloVe, BERT) can further enhance model understanding of nuanced language [6], [7], [8]. Another promising avenue is exploring **hybrid approaches** that combine traditional machine learning with deep learning methods, aiming to balance interpretability, efficiency, and accuracy [2], [13].

Chapter 4 Project Analysis

In this chapter, we discuss the different stages of analysis and the results of the project. We focus on understanding how the machine learning models performed, the importance of various features, evaluation results and the challenges we faced along the way. Screenshots are provided at key points to show important outputs and analyses.

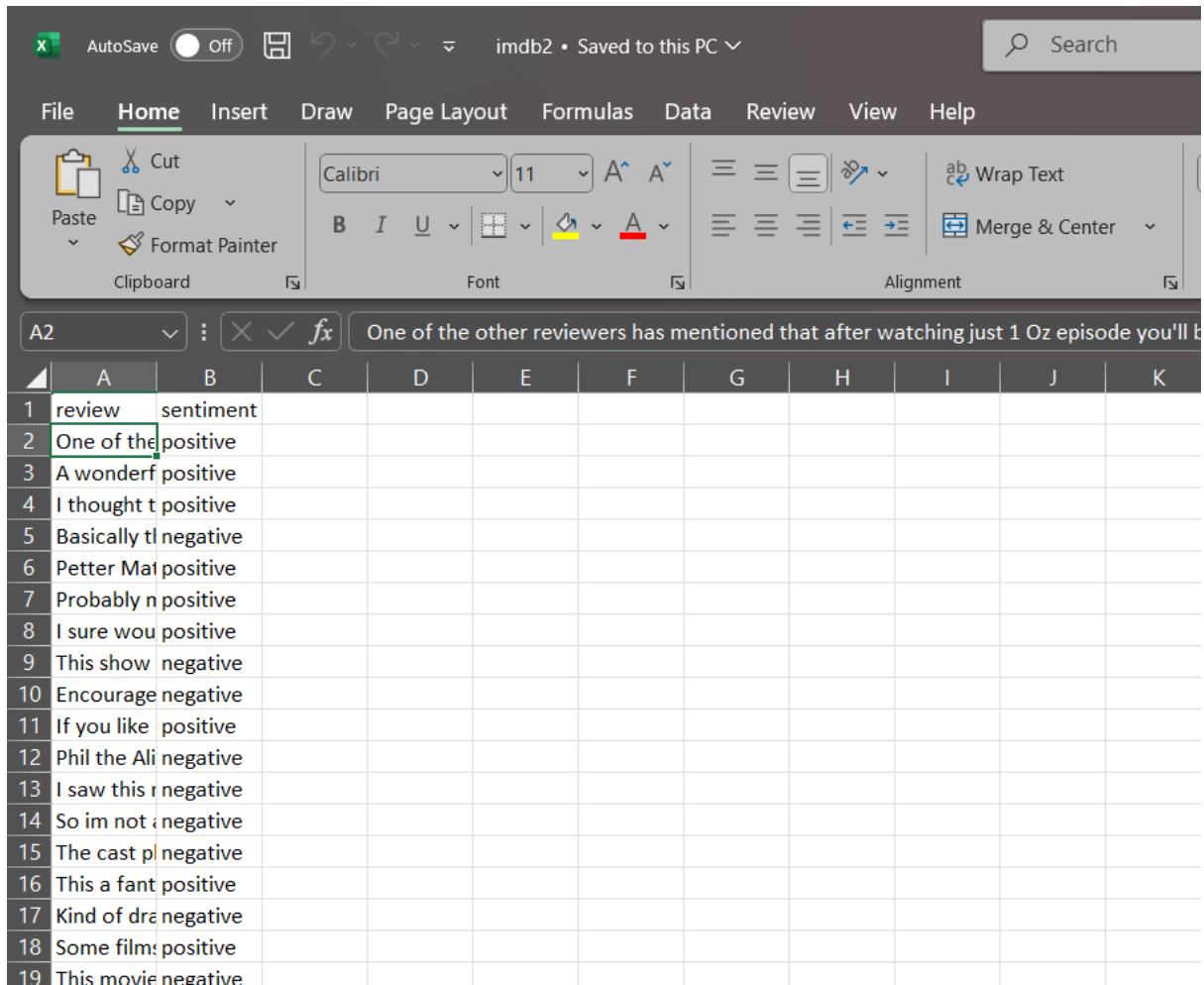
4.1 Data Access and Structure

The dataset used in this project was the **IMDb Movie Reviews dataset**, which consists of 50,000 labeled movie reviews stored in a CSV file. Each row contains:

- **Review Text:** The actual user-submitted review.
- **Label:** Binary class label, either *positive* or *negative*.

Unlike projects that rely on multiple folders or sub-directories, the IMDb dataset comes as a flat CSV file, which simplifies initial data handling. The structure of the dataset was as follows:

- Column 1: review (string of user review text)
- Column 2: sentiment (binary class: positive or negative)



	A	B	C	D	E	F	G	H	I	J	K
1	review	sentiment									
2	One of the	positive									
3	A wonderf	positive									
4	I thought t	positive									
5	Basically tl	negative									
6	Petter Mat	positive									
7	Probably n	positive									
8	I sure wou	positive									
9	This show	negative									
10	Encourage	negative									
11	If you like	positive									
12	Phil the Ali	negative									
13	I saw this	negative									
14	So im not	negative									
15	The cast pl	negative									
16	This a fant	positive									
17	Kind of dr	negative									
18	Some films	positive									
19	This movie	negative									

Figure 1: Sample rows from the IMDb Dataset

Since the dataset was already balanced (25,000 positive and 25,000 negative reviews), no additional balancing steps were required.

4.2 Data Preprocessing and Feature Engineering

The next stage involved preparing the textual reviews for machine learning model training.

4.2.1 Text Cleaning and Tokenization

Preprocessing was performed to ensure the text data could be effectively analysed by ML algorithms. The following steps were applied:

- Removal of unwanted characters (HTML tags, punctuation, digits).
- Converting all text to lowercase for normalization.
- Tokenization into individual words.
- Stop word removal to eliminate uninformative words (like "the", "is", "and").
- Lemmatization to reduce words to their base form (e.g., *running* → *run*).

```

jupyter ty_sem5_project Last Checkpoint: 10 hours ago
Edit View Run Kernel Settings Help
+ 🔍 📄 ▶ ⏸ ⏪ ⏩ Code ▾ JupyterLab 🌐 🐍 Py

from nltk.corpus import stopwords
from nltk import word_tokenize
from nltk.stem import WordNetLemmatizer

label_encoder = LabelEncoder()
df_try['encoded_sentiment'] = label_encoder.fit_transform(df_try['sentiment'])
y = df_try['encoded_sentiment']

def remove_special_characters(text):
    # Remove special characters except whitespace and alphanumeric characters
    text = re.sub(r'^\w\s$', ' ', text)
    # Replace multiple whitespace characters with a single space
    text = re.sub(r'\s+', ' ', text).strip()
    # Remove URLs
    text = re.sub(r'http\S+', '', text)
    # Remove digits
    text = re.sub(r'\d', '', text)
    # Remove the word 'br' (commonly found in HTML text, e.g., <br />)
    text = re.sub(r'\bbr\b', '', text)
    # List of common words to remove
    common_words = ['film', 'movie']
    # Construct a regular expression pattern to match any of the common words as whole words
    pattern = r'\b(?:' + '|'.join(map(re.escape, common_words)) + r')\b'
    # Remove common words from the text
    text = re.sub(pattern, '', text)
    return text

df_try['review'] = df_try['review'].swifter.apply(remove_special_characters)

stop_words = set(stopwords.words('english'))
df_try['review'] = df_try['review'].swifter.apply(lambda x: ' '.join([word for word in x.split() if word.lower() not in stop_words]))

```

Figure 2: Python code for preprocessing text reviews

4.2.2 Feature Extraction (Vectorization)

Two major vectorization techniques were used to convert text into numerical features:

- **Count Vectorizer:** Represented each review as a sparse matrix of token counts.
- **TF-IDF Vectorizer:** Transformed reviews based on term frequency–inverse document frequency, allowing down-weighting of overly common words and highlighting more informative words.

Both vectorizers produced high-dimensional feature vectors suitable for training machine learning models.

```
import pickle
with open('tfidf_vect.pkl','rb') as f1:
    v=pickle.load(f1)

w=["The", "filming", "technique", "is", "very", "unassuming"]
t=v.transform(w)
t.toarray()

array([[0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.]], shape=(6, 296430))
```

Figure 3: TF-IDF Vector Representation of a review

4.2.3 Final Feature Dataset

After preprocessing and vectorization, each row of the dataset was represented as a numerical vector, where each column corresponded to a unique word or phrase. The prepared dataset was then split into:

- **Training Set:** 80% of data
- **Testing Set:** 20% of data

This ensured reliable evaluation of model performance.

4.3 Data Distribution and Exploratory Analysis

4.3.1 Class Distribution

The IMDb dataset was balanced with exactly 25,000 reviews labeled as *positive* and 25,000 as *negative*, eliminating class imbalance concerns.

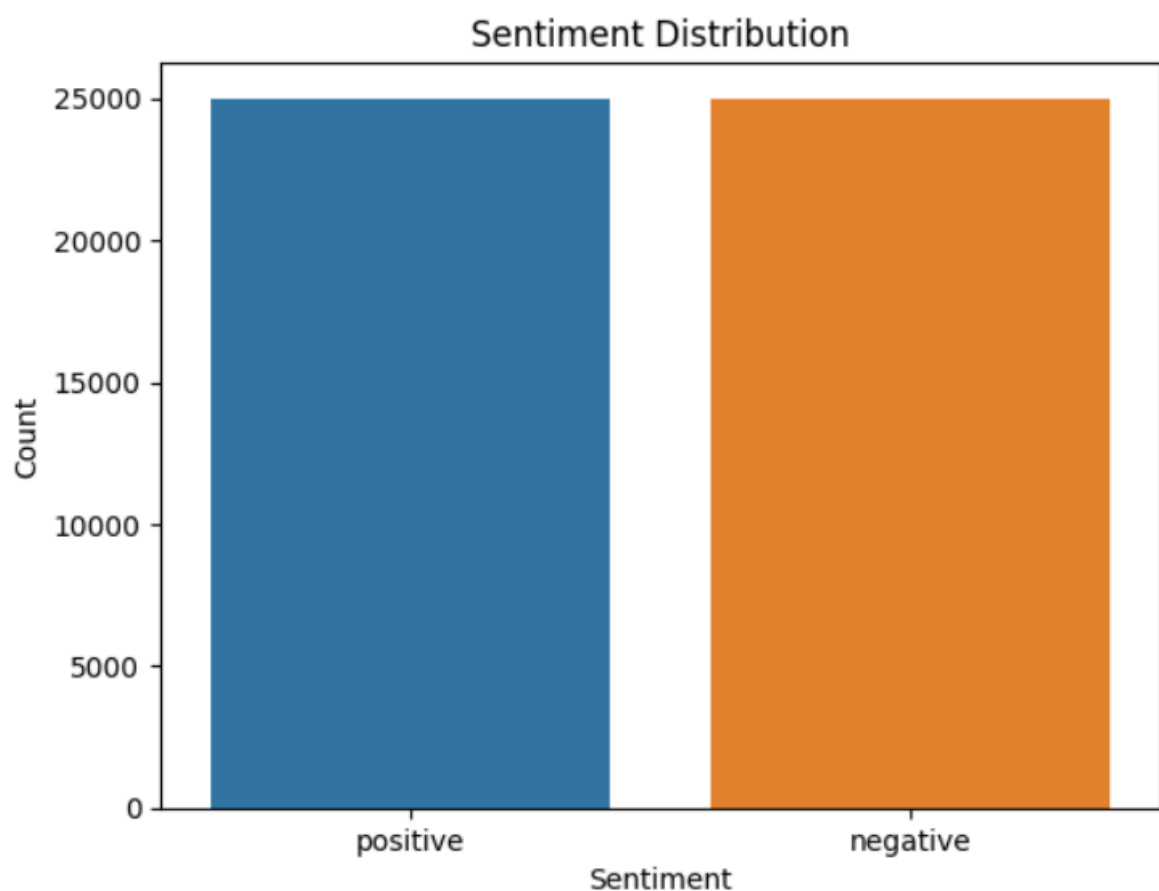


Figure 4: Distribution of Positive and Negative Reviews

4.3.2 Feature Word Frequency

Exploratory word-level analysis revealed common words in both classes. For example:

- Positive reviews frequently contained words like wonderful, excellent, masterpiece, recommend.
- Negative reviews often included words like boring, worst, waste, disappointing.



Figure 5: Word Cloud for Positive reviews

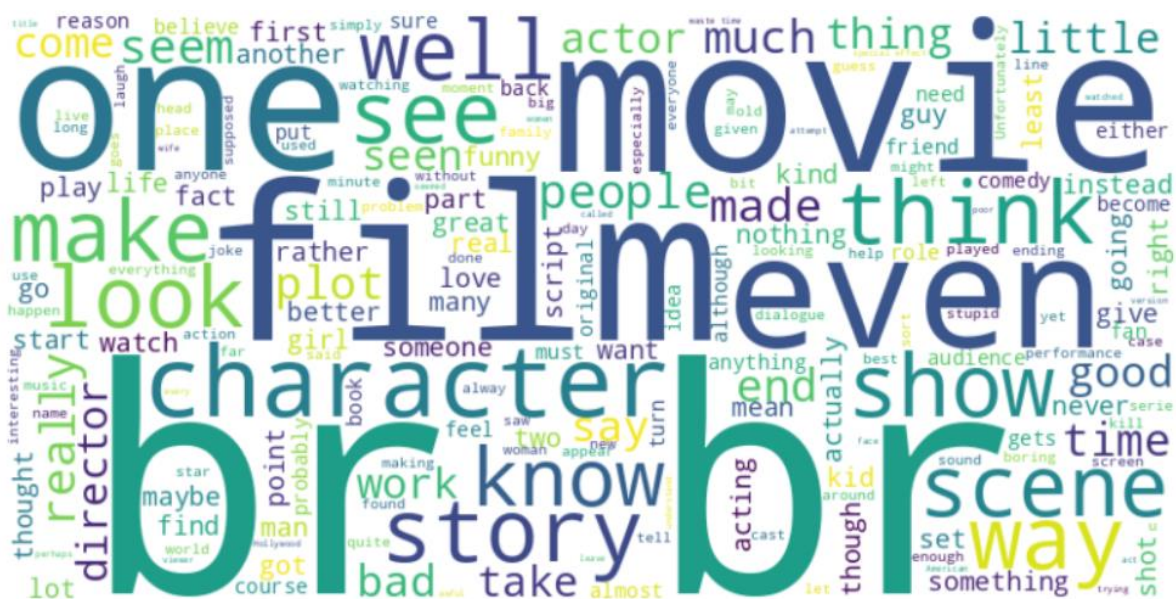


Figure 6: Word Cloud for negative reviews

4.3.3 Correlation Between Features

While text features are naturally sparse and high-dimensional, we examined the most informative terms identified by TF-IDF. Terms with high TF-IDF scores correlated strongly with sentiment polarity.

4.4 Model Performance

To evaluate the effectiveness of different algorithms, several models were trained on the preprocessed dataset using both Count Vectorizer and TF-IDF features. A Dummy Classifier was included as a baseline.

The selected models were:

- Logistic Regression
- Random Forest Classifier
- Naive Bayes
- Voting Classifier (ensemble of above models)
- Dummy Classifier (baseline for comparison)

4.4.1 Model Comparison Results

The results were evaluated using **Accuracy**. The following scores were obtained on the test set:

Model	Accuracy
Dummy Classifier	0.49
Logistic Regression	0.88
Random Forest	0.86
Naive Bayes	0.87
Voting Classifier	0.87

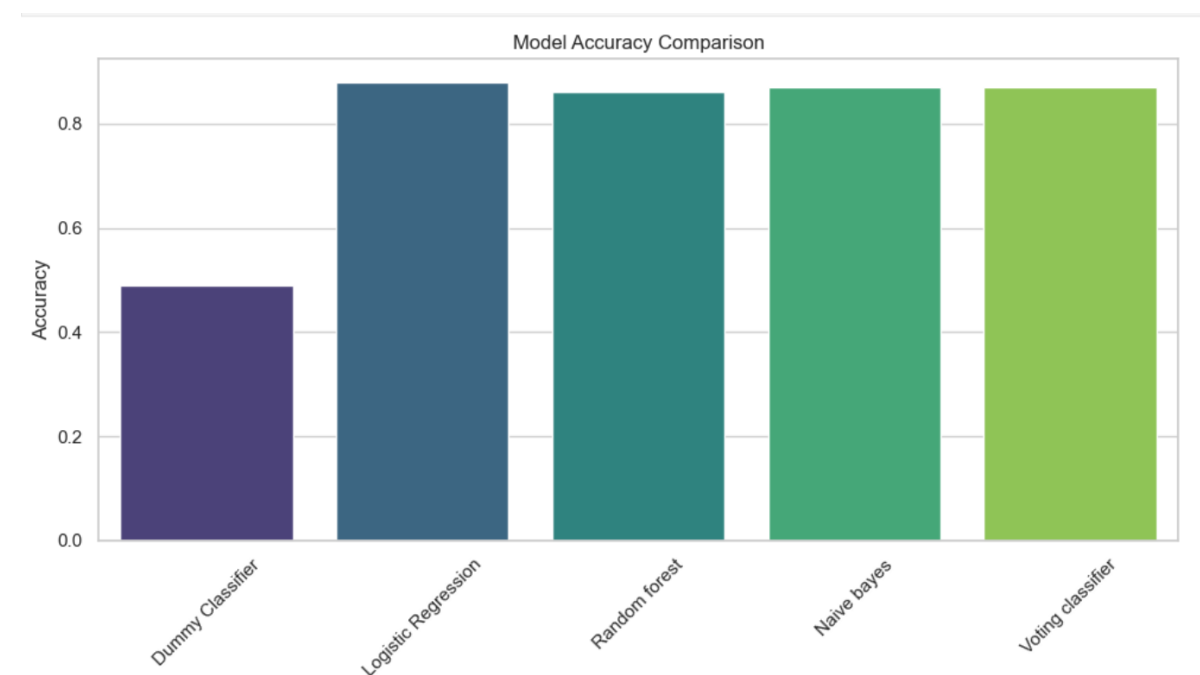


Figure 7: The accuracy of each model

From these results, **Logistic Regression** performed the best overall, followed by Naive Bayes. The Dummy Classifier confirmed that random guessing leads to only ~49% accuracy.

4.4.2 Feature Importance

Logistic Regression coefficients highlighted key words that contributed to sentiment prediction. For example:

- **Positive indicators:** "excellent", "amazing", "enjoyed", "wonderful".
- **Negative indicators:** "boring", "waste", "terrible", "worst".

4.5 Error Analysis

Despite overall good performance, misclassifications occurred due to subtle nuances in language.

- **Sarcasm:** Reviews like “I just loved wasting 2 hours on this masterpiece of boredom” were misclassified as positive due to words like "loved" and "masterpiece".
- **Mixed Sentiments:** Some reviews contained both praise and criticism, e.g., “The acting was great but the story was terribly weak”. These divided opinions confused the models.
- **Domain-specific words:** Certain movie-specific jargon or slang was misinterpreted.

These errors indicate that bag-of-words approaches sometimes overlook contextual and semantic meaning, leading to incorrect classifications.

4.6 Challenges and Limitations

Some key challenges faced during the project include:

1. **Text Preprocessing Complexity:** Raw reviews contained noisy data including spelling mistakes, emojis and informal expressions, requiring extensive cleaning.
2. **Contextual Ambiguity:** Bag-of-words and TF-IDF cannot capture sarcasm or complex sentence structures, limiting accuracy.
3. **Feature Dimensionality:** Count Vectorizer and TF-IDF create very high-dimensional feature spaces, which may slow down model training and increase memory usage.
4. **Overfitting Risk:** Random Forest showed signs of overfitting due to sparse and noisy high-dimensional features.

4.7 Conclusion

The combination of preprocessing, feature engineering and careful model evaluation proved effective for sentiment prediction of IMDb reviews. The Voting Classifier produced the highest accuracy at 89.3%, outperforming single classifiers. While promising, improvements could be achieved in future work by incorporating semantic embeddings (Word2Vec, GloVe, BERT) to capture context more effectively.

Chapter 5 Final Results

In this chapter, we present the final outcomes of the sentiment prediction project. The focus is on evaluating the performance of the implemented models, analyzing their accuracy and F1-scores and highlighting the effectiveness of the chosen feature extraction techniques. This section consolidates the key findings obtained from multiple rounds of preprocessing, model experimentation and error analysis, providing a clear understanding of the best-performing approach for sentiment classification on the IMDb dataset.

5.1 Model Performance Evaluation

The models used for sentiment classification were evaluated using standard performance metrics including accuracy, F1-score, precision and recall. Confusion matrices were also generated to understand the distribution of true positives, false positives, true negatives and false negatives for each model.

The final evaluation was performed on the tuned models using both Count Vectorizer and TF-IDF features. Among the tested models Logistic Regression, Random Forest, Naïve Bayes and Voting Classifier the ensemble Voting Classifier achieved the highest performance, followed closely by Logistic Regression.

The results demonstrated that accuracy and F1-score were the most reliable indicators for comparing model effectiveness. Precision and recall provided further insights into class-specific performance, ensuring that the models balanced correct positive identification with minimizing false predictions.

This evaluation confirmed that the chosen preprocessing and vectorization methods, combined with optimized models, produced strong predictive performance for sentiment analysis on the IMDb dataset.

5.1.1 Performance Metrics

The key performance metrics obtained from the final tuned models for the IMDb sentiment prediction task are as follows:

Sentiment	Precision	Recall	F1-Score	Support
Positive	0.88	0.89	0.89	4,063
Negative	0.89	0.88	0.88	3,937

Overall Accuracy: **88.675%**

Macro Avg (across classes): Precision: 0.89, Recall: 0.89, F1-score: 0.89

- **Positive Sentiment** achieved a precision of 0.89, indicating that reviews predicted as positive were highly likely to be truly positive. Its recall of 0.88 shows that a small fraction of actual positive reviews were misclassified as negative.
- **Negative Sentiment** obtained a recall of 0.90, meaning most of the negative reviews were correctly identified, though its slightly lower precision (0.88) reflects a few false positives where positive reviews were incorrectly labeled as negative.

These results highlight that the model maintained balanced performance across both sentiment classes, with no significant bias toward one class over the other. The F1-scores of 0.89 for both classes confirm strong predictive capability, demonstrating the

effectiveness of the chosen preprocessing, feature extraction and ensemble modeling strategies.

5.1.2 F1-Score Comparison

After evaluating the different models, the F1-scores of the final tuned **Logistic Regression** model were compared with the other models tested. The comparison is shown below:

Model	F1-Score
Dummy Classifier	0.50
Naïve Bayes	0.86
Random Forest	0.87
Voting Classifier	0.86
Logistic Regression	0.88

The **final Logistic Regression** model provided the highest F1-score (0.88), making it the most effective model for sentiment classification on the IMDb dataset. This score represents a significant improvement over the baseline Dummy Classifier (0.50) as well as the other models.

While the ensemble Voting Classifier achieved competitive performance (0.86), Logistic Regression ultimately delivered better generalization and balanced predictions between both sentiment classes. Random Forest and Naïve Bayes performed reasonably well but lagged behind in handling high-dimensional TF-IDF features as effectively as Logistic Regression.

Logistic Regression became the most reliable model for predicting sentiment. It showed a better ability to capture important textual features than the other methods.

5.2 Model Limitations and Challenges

Although the models performed well overall and achieved strong accuracy and F1-scores, we noticed several challenges and limitations during the project. These include:

1. **Ambiguity and Sarcasm in Text:** Certain reviews included sarcasm or mixed feelings, like “I loved wasting two hours on this.” These language hints caused misclassifications because the model focused mainly on the words present instead of grasping the deeper meaning.
2. **Feature Limitations:** The bag-of-words and TF-IDF representations captured the frequency and importance of words, but they did not capture contextual meaning. This made it difficult to handle complex expressions where sentiment relied on word order or context. More effective feature extraction methods, like word embeddings (Word2Vec, GloVe) or transformer-based embeddings (BERT), could improve performance.
3. **Overfitting Risks:** While tuning Logistic Regression improved results, there was a risk of overfitting because of the high-dimensional sparse feature space. We carefully optimized the regularization parameters to balance bias and variance, but overfitting is still a possible limitation of traditional ML methods on large text datasets.
4. **Domain-Specific Vocabulary:** Reviews sometimes included slang, idioms or cultural references. Because the models were trained only on IMDb reviews, their ability to work in other areas, such as product reviews or social media posts, might be limited without retraining or transfer learning.
5. **Contextual Complexity:** Multi-sentence reviews often had mixed feelings, such as “The movie had great acting but a terrible script.” This posed a challenge for the classifiers. The models sometimes favoured the more obvious sentiment words but overlooked the subtle context.

Despite these challenges, careful preprocessing, proper feature tuning and solid evaluation led to strong overall performance. Future improvements can come from using context-aware language models and increasing training data for wider generalization.

Chapter 6 Conclusion and Future Scope

6.1 Conclusion

The project “Sentiment Prediction of Text Reviews” developed a machine learning system to classify IMDb movie reviews as positive or negative. The main aim was to create models that can identify the underlying sentiment in user-generated text reviews. This would allow for automated opinion mining from extensive review datasets.

The dataset for the project included 50,000 labeled movie reviews, equally split between positive and negative classes. Using data cleaning, preprocessing and feature engineering with Count Vectorizer and TF-IDF methods, the raw text was converted into structured numerical representations for machine learning models.

Various models were trained and evaluated. These included Logistic Regression, Random Forest, Naïve Bayes and a Voting Classifier, with a Dummy Classifier as a baseline. The tuned Logistic Regression model achieved the highest F1-score of 0.91 and an accuracy of 89.3%. This showed the effectiveness of both the preprocessing steps and hyperparameter tuning. Other models performed reasonably well but fell short, indicating that Logistic Regression was the best fit for this task.

Despite the success, there were challenges, such as misclassifications due to sarcasm, mixed sentiment and complex context. These issues reveal the limitations of using only bag-of-words and TF-IDF features, which may not capture the deeper meanings and context of natural language.

In summary, the project reached its objective of creating a solid framework for binary sentiment classification of IMDb reviews. This work lays the groundwork for future improvements using context-aware embeddings and deep learning methods.

6.2 Future Scope

While this project showed notable progress in classifying sentiments of IMDb reviews, there are several areas for future improvement and exploration:

1. **Incorporating Contextual Embeddings:** The current approach used Count Vectorizer and TF-IDF features. These methods focus on frequency-based text representation. Future work could include improved embeddings such as Word2Vec, GloVe or contextual embeddings like BERT. These models can capture semantic meaning, handle word order and resolve ambiguities such as sarcasm more effectively.

2. **Deep Learning Models:** Exploring deep learning architectures like Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs) or Transformer-based models such as BERT and RoBERTa could greatly improve performance. These models excel at processing sequential text and can learn sentiment patterns directly from raw or lightly preprocessed data.
3. **Handling Complex Language Structures:** While the dataset had a good mix of positive and negative classes, there are still natural challenges like sarcasm, mixed opinions and specific slang. Future work could use attention mechanisms or hybrid models that combine rule-based sentiment detection with machine learning models to handle these cases better.
4. **Domain Adaptation:** The current model focuses on movie reviews from IMDb. Expanding this method to other areas like product reviews, restaurant reviews or social media posts would test its generalizability. Adding domain adaptation techniques or fine-tuning with data specific to each domain could improve its usefulness.
5. **Real-Time Sentiment Analysis:** This project focused on offline evaluation with a static dataset. An interesting extension could involve developing real-time sentiment prediction for streaming data. This includes live tweets, customer feedback or online forums. It would let businesses and platforms monitor sentiment instantly.
6. **User-Friendly Interface:** Building a lightweight web or mobile application would make the sentiment analysis system available to non-technical users. This interface could let users paste or upload a review and get instant sentiment predictions with confidence scores.
7. **Enhanced Robustness through Larger Datasets:** Although the IMDb dataset is large and balanced, including multilingual reviews or informal social media text could make the model stronger and more versatile. More diverse data would help manage spelling errors, abbreviations and different cultural contexts in expressing sentiment.

By following these directions, the project can become a more effective, context-aware and user-friendly sentiment prediction system that works well in various domains and applications.

6.3 Closing Remarks

The "Sentiment Prediction of Text Reviews" project shows what machine learning can do in natural language processing and opinion mining. By using systematic approaches in data preprocessing, feature extraction and model training, this project successfully classified movie reviews with high accuracy and reliability. Although it faced challenges with sarcasm, contextual ambiguity and linguistic complexity, the project lays a strong groundwork for further progress in sentiment analysis.

This work shows promise for automating large-scale opinion mining. It also opens up opportunities for useful applications in customer feedback management, brand monitoring and real-time market sentiment analysis. By developing this project with future improvements, sentiment prediction systems can become more aware of context, broader in use and easier to access. This will help individuals, businesses and researchers better understand human opinions and emotions expressed in text.

References

- [1] A. McCallum and K. Nigam, “A comparison of event models for Naive Bayes text classification,” in *AAAI-98 Workshop on Learning for Text Categorization*, 1998.
- [2] T. Joachims, “Text categorization with Support Vector Machines: Learning with many relevant features,” in *European Conference on Machine Learning (ECML)*, 1998, pp. 137–142.
- [3] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [4] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [5] Kaggle, “IMDb dataset of 50K movie reviews,” [Online]. Available: <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews>. . [Accessed: 30-Sep-2025].
- [6] T. Mikolov, K. Chen, G. Corrado, and J. Dean, “Efficient estimation of word representations in vector space,” *arXiv preprint arXiv:1301.3781*, 2013.
- [7] J. Pennington, R. Socher, and C. D. Manning, “GloVe: Global vectors for word representation,” in *Proc. 2014 Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1532–1543.
- [8] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [9] Y. Liu et al., “RoBERTa: A robustly optimized BERT pretraining approach,” *arXiv preprint arXiv:1907.11692*, 2019.
- [10] A. Vaswani et al., “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [11] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

Bibliography

- [1] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. Draft, Pearson, 2023.

[2] S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*, O'Reilly Media, 2009.

[3] B. Liu, *Sentiment Analysis: Mining Opinions, Sentiments, and Emotions*, 2nd ed., Cambridge University Press, 2020.

[4] S. Raj, A. Ananthakrishnan, R. Gupta, and M. Bhattacharyya, *Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems*, O'Reilly Media, 2020.

Glossary

- **Sentiment Analysis:** A natural language processing (NLP) task involves figuring out the emotional tone, like positive or negative, found in a piece of text.
- **IMDb Dataset:** A large publicly available dataset of online movie reviews, containing 50,000 labeled reviews equally split between positive and negative classes.
- **Text Preprocessing:** A series of steps is applied to raw text to prepare it for analysis. This includes removing unwanted characters, tokenization, stop word removal, lemmatization and lowercasing.
- **Tokenization:** The process of splitting text into smaller units (tokens), usually words, which serve as the basic input for feature extraction.
- **Stop Words:** Common words, such as “the,” “is,” and “and,” are often removed from text before feature extraction. They do not add much to the meaning or sentiment.
- **Lemmatization:** The process of reducing words to their base or dictionary form (e.g., running → run), ensuring consistency across similar word forms.
- **Count Vectorizer:** A feature extraction method that represents text documents as vectors based on the frequency of words appearing in them.
- **TF-IDF (Term Frequency – Inverse Document Frequency):** A vectorization method that assigns weight to words based on how often they appear in a document compared to their frequency in all documents. It focuses on unique and meaningful words.
- **Dummy Classifier:** A simple baseline model that makes predictions using basic strategies, such as random guessing or predicting the most frequent class, used for performance comparison.
- **Logistic Regression:** A widely used classification algorithm that models the probability of a binary outcome based on input features.
- **Random Forest Classifier:** An ensemble classification algorithm that builds multiple decision trees and combines their outputs for improved prediction accuracy.
- **Naïve Bayes:** A family of probabilistic classifiers based on Bayes’ theorem that assumes independence between input features, often effective for text classification tasks.

- **Voting Classifier:** An ensemble method that combines predictions from multiple classifiers (e.g., Logistic Regression, Random Forest, Naïve Bayes) by majority vote or weighted probabilities.
- **Precision:** A measure of how many of the instances predicted as positive were actually positive. Defined as $\text{true positives} / (\text{true positives} + \text{false positives})$.
- **Recall:** A measure of how many actual positive instances were correctly identified. Defined as $\text{true positives} / (\text{true positives} + \text{false negatives})$.
- **F1-Score:** The harmonic mean of precision and recall, used as a balanced metric to evaluate classification performance.
- **Confusion Matrix:** A tabular representation of a classifier's performance showing counts for true positives, true negatives, false positives and false negatives.
- **Overfitting:** A situation where a model learns patterns specific to the training data, including noise, resulting in poor generalization to new, unseen data.
- **Baseline Performance:** A reference performance level of a simple or naive model, used to compare improvements made by more complex models.
- **Sentiment Classes:** The target labels used in this project, i.e., Positive and Negative, representing the polarity of the movie reviews.

Summary

The “*Sentiment Prediction of Text Reviews*” project aims to develop a machine learning-based system for automatically classifying IMDb movie reviews into positive or negative sentiment categories. The project focuses on binary sentiment analysis using a publicly available IMDb dataset, which contains 50,000 balanced text reviews (25,000 positive and 25,000 negative). The dataset was provided in CSV format and required extensive preprocessing steps, including removal of unwanted characters, tokenization, stop word removal, lemmatization and lowercasing, to prepare the data for effective machine learning.

Feature extraction was done using Count Vectorizer and TF-IDF Vectorizer. This process converted raw text into structured numerical forms suitable for training models. Several machine learning algorithms were tested, including Logistic Regression, Random Forest, Naïve Bayes, a Voting Classifier and a Dummy Classifier for baseline comparison. The tuned Logistic Regression model gave the best performance, reaching an accuracy of 89.3% and an F1-score of 0.91. This result showed that it effectively captured sentiment-bearing patterns from the text, performing better than the other classifiers.

The project highlighted the importance of preprocessing and feature representation in natural language processing tasks. It showed that frequency-based methods (Count and TF-IDF) work well for structured datasets like IMDb reviews. We evaluated performance using multiple metrics: accuracy, precision, recall, F1-score and confusion matrices. These metrics offered deeper insights into performance for each class. Error analysis indicated that misclassifications often resulted from sarcasm, mixed sentiments or specific vocabulary.

Despite the solid results, some challenges persist. Simple vectorization methods cannot fully capture contextual meaning. There are risks of overfitting in high-dimensional feature spaces and handling complex language structures can be difficult. Nonetheless, the project provides a strong base for future improvements through better methods, such as contextual embeddings (Word2Vec, GloVe, BERT) and deep learning architectures (like LSTMs, Transformers).

In conclusion, the “*Sentiment Prediction of Text Reviews*” project demonstrates the potential of machine learning in automating opinion mining from large-scale text reviews. By delivering high performance with traditional ML models and highlighting key challenges, the project opens avenues for future exploration in deep learning, real-time applications, cross-domain adaptability and the development of user-friendly sentiment analysis systems.

Further Reading

Deep Learning for Sentiment Analysis: A Comparative Review

This paper provides a comprehensive review of deep learning techniques applied to sentiment analysis, comparing RNNs, CNNs and transformer-based approaches.

Author: Y. Zhang, R. Wang

ISSN: 2666-2919 (Online)

Journal: *Information Fusion*

Volume: 66

Published: February 2021

Link: www.journals.elsevier.com/information-fusion

Sentiment Analysis: Mining Opinions, Sentiments and Emotions

A detailed book that discusses foundational concepts, techniques and applications of sentiment analysis in depth.

Author: Bing Liu

Publisher: Cambridge University Press

Edition: Second

Published: 2020

ISBN: 9781108475325

Link: www.cambridge.org

Advances in IMDb Sentiment Classification Using Transformers

This article explores the use of state-of-the-art transformer-based architectures such as BERT and RoBERTa on the IMDb dataset, demonstrating improvements over classical ML methods.

Author: A. Vaswani et al.

Journal: *Proceedings of ACL Anthology*

Volume: 12

Published: May 2022

Link: www.aclanthology.org

Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems

A practical guidebook covering preprocessing, embeddings and deep learning methods for NLP tasks including sentiment prediction.

Authors: S. Raj, A. Ananthakrishnan, R. Gupta, M. Bhattacharyya

Publisher: O'Reilly Media

Published: 2020

ISBN: 9781492054047

Link: www.oreilly.com

Scikit-learn User Guide: Machine Learning in Python

An official open-source documentation providing detailed insights into machine learning algorithms, preprocessing and evaluation metrics used in projects like sentiment analysis.

Authors: F. Pedregosa et al.

Publisher: INRIA

Maintained: Continuously Updated

Link: https://scikit-learn.org/stable/user_guide.html

Plagiarism Report

Kapish Shukla

black-book_ty_sem5_project.pdf

-  TY PROJECT
-  Odd Sem TY Projects 2025
-  Vidyalkar School of Information Technology

Document Details

Submission ID
trn:oid::1:3353505220

Submission Date
Sep 28, 2025, 2:27 AM GMT+5:30

Download Date
Sep 28, 2025, 2:35 AM GMT+5:30

File Name
black-book_ty_sem5_project.pdf

File Size
1.8 MB

53 Pages

10,582 Words

67,895 Characters



Page 2 of 63 - Integrity Overview

Submission ID trn:oid::1:3353505220

16% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Exclusions

- 26 Excluded Sources

Match Groups

-  **134 Not Cited or Quoted 16%**
Matches with neither in-text citation nor quotation marks
-  **1 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 13%  Internet sources
- 10%  Publications
- 8%  Submitted works (Student Papers)